

Problem 1

Use differentials to show that, for large n and small a , $\sqrt{n+a} - \sqrt{n} \cong \frac{a}{2\sqrt{n}}$.

Find the approximate value of $\sqrt{10^{26}+5} - \sqrt{10^{26}}$.

$$\sqrt{10^{26}+5} - \sqrt{10^{26}} \cong \frac{5}{2\sqrt{10^{26}}} = 2.5 \times 10^{-13}$$

Problem 2

(1)

Given $z = (x+y)^5$, $y = \sin 10x$, find dz/dx

(2)

Given $c = \sin(a-b)$, $b = ae^{2a}$, find dc/da .

$$(1) \quad \frac{dz}{dx} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} \frac{dy}{dx} = 5(x+y)^4 (1+10\cos 10x)$$

$$(2) \quad \frac{dc}{da} = \frac{\partial c}{\partial a} + \frac{\partial c}{\partial b} \frac{db}{da} = (1-2b-e^{2a})\cos(a-b)$$

Problem 3

(1)

If $z = xe^{-y}$ and $x = \cosh t$, $y = \cos s$, find $\partial z/\partial s$ and $\partial z/\partial t$.

(2)

If $w = e^{-r^2-s^2}$, $r = uv$, $s = u+2v$, find $\partial w/\partial u$ and $\partial w/\partial v$.

(3)

If $w = x+y$ with $x^3+xy+y^3 = s$ and $x^2y+xy^2 = t$, find $\partial w/\partial s$, $\partial w/\partial t$.

$$(1) \quad \frac{\partial z}{\partial s} = \frac{\partial z}{\partial y} \frac{dy}{ds} = xe^{-y} \sin s = z \sin s, \quad \frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{dx}{dt} = e^{-y} \sinh t$$

$$(2) \quad \frac{\partial w}{\partial u} = \frac{\partial w}{\partial r} \frac{\partial r}{\partial u} + \frac{\partial w}{\partial s} \frac{\partial s}{\partial u} = -2(rv+s)w, \quad \frac{\partial w}{\partial v} = \frac{\partial w}{\partial r} \frac{\partial r}{\partial v} + \frac{\partial w}{\partial s} \frac{\partial s}{\partial v} = -2(ru+2s)w$$

$$(3) \quad (x+y)^3 + xy = s+3t, \quad xy = \frac{t}{w}$$

$$\text{故, } w^3 + \frac{t}{w} = s+3t$$

$$\text{两边对 } s \text{ 求导: } 3w^2 \frac{\partial w}{\partial s} - \frac{t}{w^2} \frac{\partial w}{\partial s} = 1, \quad 3w^3 \frac{\partial w}{\partial s} - xy \frac{\partial w}{\partial s} = w, \quad \frac{\partial w}{\partial s} = \frac{w}{3w^3 - xy}$$

$$\text{同理, 两边对 } t \text{ 求导可得: } \frac{\partial w}{\partial t} = \frac{3w-1}{3w^3 - xy}$$

Problem 4

Find the point on $2x + 3y + z - 11 = 0$ for which $4x^2 + y^2 + z^2$ is a minimum.

$$\text{令 } f(x, y, z, \lambda) = 4x^2 + y^2 + z^2 - \lambda(2x + 3y + z - 11)$$

$$\text{令 } \frac{\partial f}{\partial x} = 8x - 2\lambda = 0, \quad \frac{\partial f}{\partial y} = 2y - 3\lambda = 0, \quad \frac{\partial f}{\partial z} = 2z - \lambda = 0,$$

$$\frac{\partial f}{\partial \lambda} = -(2x + 3y + z - 11) = -\left(\frac{\lambda}{2} + \frac{9\lambda}{2} + \frac{\lambda}{2} - 11\right) = 0$$

$$\text{则 } \lambda = 2, \quad x = \frac{1}{2}, \quad y = 3, \quad \lambda_1 = 1$$

$$\text{则 } P\left(\frac{1}{2}, 3, 1\right)$$

Problem 5

Find the shortest distance from the origin to the line of intersection of the planes $2x + y - z = 1$ and $x - y + z = 2$.

$$\text{令 } f(x, y, z, \lambda_1, \lambda_2) = x^2 + y^2 + z^2 - \lambda_1(2x + y - z - 1) - \lambda_2(x - y + z - 2)$$

$$\text{令 } \frac{\partial f}{\partial x} = 2x - 2\lambda_1 - \lambda_2 = 0, \quad \frac{\partial f}{\partial y} = 2y - \lambda_1 + \lambda_2 = 0, \quad \frac{\partial f}{\partial z} = 2z + \lambda_1 - \lambda_2 = 0,$$

$$\frac{\partial f}{\partial \lambda_1} = -(2x + y - z - 1) = -\left(2\lambda_1 + \lambda_2 + \frac{\lambda_1 - \lambda_2}{2} - \frac{\lambda_2 - \lambda_1}{2} - 1\right) = 0,$$

$$\frac{\partial f}{\partial \lambda_1} = -(x - y + z - 2) = -\left(\frac{2\lambda_1 + \lambda_2}{2} - \frac{\lambda_1 - \lambda_2}{2} + \frac{\lambda_2 - \lambda_1}{2} - 2\right) = 0$$

$$\text{解得, } \lambda_1 = \frac{1}{3}, \lambda_2 = \frac{4}{3}, x=1, y=-\frac{1}{2}, z=\frac{1}{2}$$

$$\text{则 } d = \sqrt{1^2 + \left(-\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \frac{\sqrt{6}}{2}$$

Problem 6

In the partial differential equation

$$\frac{\partial^2 z}{\partial x^2} - 5 \frac{\partial^2 z}{\partial x \partial y} + 6 \frac{\partial^2 z}{\partial y^2} = 0$$

put $s = y + 2x$, $t = y + 3x$ and show that the equation becomes $\partial^2 z / \partial s \partial t = 0$. Following the method of solving (11.6), solve the equation.

$$(11.6) \quad \frac{\partial^2 F}{\partial x^2} - \frac{1}{v^2} \frac{\partial^2 F}{\partial t^2} = 4 \frac{\partial^2 F}{\partial r \partial s} = 0.$$

We can easily solve (11.6). We have

$$\frac{\partial^2 F}{\partial r \partial s} = \frac{\partial}{\partial r} \left(\frac{\partial F}{\partial s} \right) = 0,$$

that is, the r derivative of $\partial F / \partial s$ is zero. Then $\partial F / \partial s$ must be independent of r , so $\partial F / \partial s =$ some function of s alone. We integrate with respect to s to find $F = f(s) + \text{"const."}$; the "constant" is a constant as far as s is concerned, but it may be any function of r , say $g(r)$, since $(\partial / \partial s)g(r) = 0$. Thus we find that the solution of (11.6) is

$$(11.7) \quad F = f(s) + g(r).$$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial s} \frac{\partial s}{\partial x} + \frac{\partial z}{\partial t} \frac{\partial t}{\partial x} = 2 \frac{\partial z}{\partial s} + 3 \frac{\partial z}{\partial t} = \left(2 \frac{\partial}{\partial s} + 3 \frac{\partial}{\partial t} \right) z,$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial s} \frac{\partial s}{\partial y} + \frac{\partial z}{\partial t} \frac{\partial t}{\partial y} = \frac{\partial z}{\partial s} + \frac{\partial z}{\partial t} = \left(\frac{\partial}{\partial s} + \frac{\partial}{\partial t} \right) z$$

$$\text{则有, } \frac{\partial}{\partial x} = \left(2 \frac{\partial}{\partial s} + 3 \frac{\partial}{\partial t} \right), \quad \frac{\partial}{\partial y} = \left(\frac{\partial}{\partial s} + \frac{\partial}{\partial t} \right)$$

$$\text{则, } \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = \left(2 \frac{\partial}{\partial s} + 3 \frac{\partial}{\partial t} \right) \left(2 \frac{\partial z}{\partial s} + 3 \frac{\partial z}{\partial t} \right) = 4 \frac{\partial^2 z}{\partial s^2} + 12 \frac{\partial^2 z}{\partial s \partial t} + 9 \frac{\partial^2 z}{\partial t^2},$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right) = \left(\frac{\partial}{\partial s} + \frac{\partial}{\partial t} \right) \left(\frac{\partial z}{\partial s} + \frac{\partial z}{\partial t} \right) = \frac{\partial^2 z}{\partial s^2} + \frac{\partial^2 z}{\partial s \partial t} + \frac{\partial^2 z}{\partial t^2},$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) = \left(2 \frac{\partial}{\partial s} + 3 \frac{\partial}{\partial t} \right) \left(\frac{\partial z}{\partial s} + \frac{\partial z}{\partial t} \right) = 2 \frac{\partial^2 z}{\partial s^2} + 5 \frac{\partial^2 z}{\partial s \partial t} + 3 \frac{\partial^2 z}{\partial t^2}$$

$$\text{则, } \frac{\partial^2 z}{\partial x^2} - 5 \frac{\partial^2 z}{\partial x \partial y} + 6 \frac{\partial^2 z}{\partial y^2} = -7 \frac{\partial^2 z}{\partial s \partial t} = 0$$

$$\text{则, } \frac{\partial^2 z}{\partial s \partial t} = 0 \text{ 得证}$$

The solution is: $z = f(s) + g(t) = f(y + 2x) + g(y + 3x)$

Problem 7

Suppose that $w = f(x, y)$ satisfies

$$\frac{\partial^2 w}{\partial x^2} - \frac{\partial^2 w}{\partial y^2} = 1.$$

Put $x = u + v$, $y = u - v$, and show that w satisfies $\partial^2 w / \partial u \partial v = 1$. Hence solve the equation.

$$u = \frac{x+y}{2}, \quad v = \frac{x-y}{2}$$

$$\frac{\partial w}{\partial x} = \frac{\partial w}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial w}{\partial v} \frac{\partial v}{\partial x} = \frac{1}{2} \left(\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right) w, \quad \frac{\partial w}{\partial y} = \frac{\partial w}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial w}{\partial v} \frac{\partial v}{\partial y} = \frac{1}{2} \left(\frac{\partial}{\partial u} - \frac{\partial}{\partial v} \right) w$$

$$\frac{\partial^2 w}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial w}{\partial x} \right) = \frac{1}{4} \left(\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right)^2 w = \frac{1}{4} \left(\frac{\partial^2}{\partial u^2} + 2 \frac{\partial^2}{\partial u \partial v} + \frac{\partial^2}{\partial v^2} \right) w,$$

$$\frac{\partial^2 w}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial w}{\partial y} \right) = \frac{1}{4} \left(\frac{\partial}{\partial u} - \frac{\partial}{\partial v} \right)^2 w = \frac{1}{4} \left(\frac{\partial^2}{\partial u^2} - 2 \frac{\partial^2}{\partial u \partial v} + \frac{\partial^2}{\partial v^2} \right) w$$

$$\frac{\partial^2 w}{\partial x^2} - \frac{\partial^2 w}{\partial y^2} = \frac{1}{4} \left(\frac{\partial^2}{\partial u^2} + 2 \frac{\partial^2}{\partial u \partial v} + \frac{\partial^2}{\partial v^2} \right) w - \frac{1}{4} \left(\frac{\partial^2}{\partial u^2} - 2 \frac{\partial^2}{\partial u \partial v} + \frac{\partial^2}{\partial v^2} \right) w = \frac{\partial^2 w}{\partial u \partial v} = 1 \text{ 得证}$$

$$\text{The solution is: } w = \frac{x^2 - y^2}{4} + F(x + y) + G(x - y)$$

Problem 8

For each of the following differential equations, separate variables and find a solution containing one arbitrary constant. Then find the value of the constant to give a particular solution satisfying the given boundary condition. Computer plot a slope field and some of the solution curves.

(1)

$$y' \sin x = y \ln y,$$

$$y = e \text{ when } x = \pi/3.$$

(2)

$$y dy + (xy^2 - 8x) dx = 0,$$

$$y = 3 \text{ when } x = 1.$$

$$(1) \quad \frac{dy}{dx} \sin x = y \ln y, \text{ 则 } \frac{dy}{y \ln y} = \frac{dx}{\sin x}$$

$$\text{两边积分有, } \int \frac{dy}{y \ln y} = \int \frac{dx}{\sin x}$$

$$\text{则, } y = e^{C(\csc x - \cot x)}, \quad C = \sqrt{3}$$

$$(2) \quad \frac{y}{y^2 - 8} dy = -x dx \quad (y^2 - 8 \neq 0)$$

$$\text{两边积分有, } \int \frac{y}{y^2 - 8} dy = \int -x dx$$

$$\text{即, } \frac{1}{2} \int \frac{1}{y^2 - 8} dy^2 = \int -x dx, \quad \frac{1}{2} \ln(y^2 - 8) = -\frac{1}{2} x^2 + C, \quad \ln(y^2 - 8) = C - x^2$$

$$\text{则, } y^2 = 8 + e^{C - x^2}, \quad C = 1$$

Problem 9

(1)

$$y'' - y' - 2y = 3e^{2x}$$

(2)

$$5y'' + 12y' + 20y = 120 \sin 2x$$

(3)

$$(D^2 + 2D + 17)y = 60e^{-4x} \sin 5x \quad \text{Hint: First solve } (D^2 + 2D + 17)y = 60e^{(-4+5i)x}.$$

(4)

$$5y'' + 6y' + 2y = x^2 + 6x$$

$$(1) \quad (D-2)(D+1)y = 3e^{2x}$$

$$y = Ae^{-x} + Be^{2x} + xe^{2x}$$

$$(2) \quad y = e^{-\frac{6x}{5}} \left(A \sin \frac{8x}{5} + B \cos \frac{8x}{5} \right) - 5 \cos 2x$$

$$(3) \quad y = e^{-x} (A \sin 4x + B \cos 4x) + 2e^{-4x} \cos 5x$$

$$(4) \quad y = e^{-\frac{3x}{5}} \left(A \sin \frac{x}{5} + B \cos \frac{x}{5} \right) + \frac{x^2 - 5}{2}$$

Problem 10

Solve the given differential equation by using the principle of superposition [see the solution of equation (6.29)].

$$(D-1)^2 y = 4e^x + (1-x)(e^{2x} - 1)$$

$$y = (A+Bx)e^x + 2x^2e^x + (3-x)e^{2x} + x + 1$$